Close additive-guarantee gap for deterministic impartial selection in plurality setting

ChatGPT (`gpt-5.6-sol`)*

Partial progress generated on 2026-08-24

Abstract

This note records *partial progress, not a complete solution*: the general asymptotic gap remains unresolved. The universal $\sqrt{2n} + O(1)$ upper bound is valid, and one previously unresolved finite case can be closed:

$$\boxed{L_7^* = 2.}$$

The cases $n = 8, 9$ and the order of growth for large n remain open.

1 The open problem

This paper addresses an open problem posed by Javier Cembrano, Felix Fischer, David W. Hannon, Max Klimm in *Impartial Selection with Additive Guarantees via Iterated Deletion* [1] (Economics and Computation (EC), 2022), Page 20, end of Section 3 (after the proof of Theorem 1).. The website catalogue entry is problem 21 (`W4285089261_p0`) of the Open Problems in Operations Research collection.

> *For the plurality class $\mathcal{G}_n(1)$, narrow the quantitative gap between: 1) the lower bound α_n on the worst-case additive loss of any deterministic impartial selection mechanism, and 2) the upper bound on worst-case additive loss achieved by the (optimal) twin threshold mechanism (in particular, the explicit $\sqrt{7.25\,n}$ upper bound).*
>
> *Concretely, improve the known lower bound α_n and/or improve the known upper bound for deterministic impartial mechanisms in the plurality setting so that these bounds are closer (as functions of n) under the additive-loss metric $\sup_{G\in\mathcal{G}_n(1)}(\Delta(G) - \delta^-(f(G), G))$.*

The remainder of this document is the partial progress produced by the model, reproduced verbatim from the pipeline output.

---

*Generated by the `base_solver_ni_two_iterations` pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve → A.8 verify → A.10 repair write-up; run `run_20260824_071029_a9e80c0f`, `ec-2it-ultra`); two-iteration run, outcome from iteration 2. Independent verifier assessment: REJECTED with progress score 2/3 (major partial progress). Maintained by the AI in Mathematics & Operations Research project.

2 Definitions

Let $N = [n]$. The ballot of vertex i is

$$x_i \in (N \setminus \{i\}) \cup \{\perp\},$$

where $x_i = \perp$ denotes abstention. Put

$$d_v(x) = |\{i \in N : x_i = v\}|, \qquad \Delta(x) = \max_{v \in N} d_v(x),$$

and set $d_\perp(x) = 0$.

A deterministic mechanism is a map f assigning to each profile x a value $f(x) \in N \cup \{\perp\}$. It is *impartial* if, for every i, whether $f(x) = i$ is independent of x_i. Define

$$L_n^* = \min_{\substack{f \text{ deterministic} \\ f \text{ impartial}}} \max_x \bigl(\Delta(x) - d_{f(x)}(x)\bigr).$$

The minimum exists because both the space of profiles and the space of mechanisms are finite.

3 A universal upper bound

Let

$$r = \min\left\{k \geq 1 : \frac{k(k+1)}{2} \geq n\right\}, \qquad t_j = r - j + 1 \quad (1 \leq j \leq r).$$

For nonempty $C \subseteq N$, define

$$e_C(u) = |\{i \in N \setminus C : x_i = u\}|.$$

A set C with $j = |C| \leq r$ is *terminal* if

$$e_C(c) \geq t_j \quad (c \in C), \qquad e_C(u) \leq t_j - 2 \quad (u \notin C). \tag{1}$$

If a terminal set exists, the mechanism selects, within the inclusion-maximal terminal set C^*, the maximizer of e_{C^*} under a fixed priority order; otherwise it abstains.

3.1 Terminal sets are nested

Let C and D be terminal sets with $j = |C| \leq k = |D|$, and suppose they are incomparable. Write

$$A = C \setminus D, \qquad B = D \setminus C, \qquad |A| = a \geq 1, \qquad |B| = a + k - j.$$

For $c \in A$,

$$e_C(c) - e_D(c) \geq t_j - (t_k - 2) = k - j + 2.$$

Only voters in B can contribute positively to the sum of these differences over A, and each such voter casts at most one nomination. Hence

$$a(k - j + 2) \leq \sum_{c \in A} \bigl(e_C(c) - e_D(c)\bigr) \leq |B| = a + k - j.$$

But

$$a(k - j + 2) - (a + k - j) = (a - 1)(k - j) + a > 0,$$

a contradiction. Terminal sets therefore form a chain, and C^* is unique.

3.2 Impartiality

Suppose that w is selected and that only x_w changes. Since $w \in C^*$, every value $e_{C^*}(u)$ ignores x_w; thus C^* remains terminal, and its maximizer under the fixed priority order remains w.

Any newly terminal set D is comparable with C^*. A proper subset of C^* cannot be maximal. If $D \supsetneq C^*$, then $w \in D$, so the terminality of D likewise ignores x_w; hence D was already terminal before the change, contradicting the maximality of C^*. Thus w remains selected. Applying the same argument in the reverse direction proves impartiality.

3.3 Existence of terminal sets when $\Delta \geq r$

Assume $\Delta \geq r$. Choose v with $d_v = \Delta$ and put $C_1 = \{v\}$. Because self-nominations are forbidden,

$$e_{C_1}(v) = d_v \geq r = t_1.$$

Suppose that C_j satisfies the member inequalities in (1) but is not terminal. Then some $u \notin C_j$ satisfies

$$e_{C_j}(u) \geq t_j - 1 = r - j = t_{j+1}.$$

Set $C_{j+1} = C_j \cup \{u\}$. The scores of the old members fall by at most one, while

$$e_{C_{j+1}}(u) = e_{C_j}(u),$$

so all member inequalities for C_{j+1} hold.

Let

$$S_j = \sum_{c \in C_j} e_{C_j}(c).$$

Adding u gives

$$S_{j+1} = S_j + e_{C_j}(u) - \mathbf{1}_{\{x_u \in C_j\}} \geq S_j + r - j - 1.$$

If size r were reached, then

$$S_r \geq r + \sum_{j=1}^{r-1}(r - j - 1) = \frac{r(r-1)}{2} + 1.$$

Every edge counted by S_r has its tail outside C_r, so

$$S_r \leq n - r \leq \frac{r(r-1)}{2},$$

where the last inequality follows from $n \leq r(r+1)/2$. This is impossible. A terminal set is therefore encountered whenever $\Delta \geq r$; consequently, abstention implies

$$\Delta \leq r - 1. \tag{2}$$

3.4 Loss upon selection

Let $j = |C^*|$, and let w be the selected vertex. For $c \in C^*$,

$$d_c \leq e_{C^*}(c) + j - 1 \leq e_{C^*}(w) + j - 1 \leq d_w + j - 1.$$

For $u \notin C^*$,

$$d_u \le e_{C^*}(u) + j \le t_j - 2 + j = r - 1,$$

whereas

$$d_w \ge e_{C^*}(w) \ge t_j = r - j + 1.$$

Thus $d_u - d_w \le j - 2$, while members beat w by at most $j - 1$. Together with (2),

$$\Delta - d_{f(x)} \le r - 1.$$

Therefore

$$\boxed{L_n^* \le r - 1 = \left\lceil \frac{\sqrt{8n+1} - 3}{2} \right\rceil = \sqrt{2n} + O(1).} \tag{3}$$

This is the exact worst-case loss of this mechanism. Take $r - 1$ distinct vertices nominating a common center z, with every other vertex abstaining. A set excluding z has a member of external score $0 < t_j$; the singleton $\{z\}$ has score $r - 1 < t_1 = r$; and any larger set containing z has a non-center member of score zero. Hence there is no terminal set, and the mechanism abstains, incurring loss $r - 1$.

4 Closing the case $n = 7$

For seven vertices we use the following baseline terminal sets. A set C of size $j \in \{1, 2\}$ is terminal if

$$e_C(c) \ge 4 - j \quad (c \in C), \qquad e_C(u) \le 2 - j \quad (u \notin C). \tag{4}$$

These terminal sets are nested by the argument of Section 3.1. If such a set exists, the mechanism selects the external-score maximizer in the unique maximal one. This baseline rule is impartial.

Whenever it selects, its loss is at most one: members differ from the winner by at most $j - 1 \le 1$, while every outsider satisfies

$$d_u \le e_C(u) + j \le 2$$

and the winner has degree at least $4 - j \ge 2$.

4.1 Exceptional candidates

Call w *exceptional* if the other six vertices can be named a, c, p, q, ρ, σ so that

$$x_p = x_q = w, \qquad x_c = x_\rho = a, \qquad x_\sigma = c. \tag{5}$$

There is no restriction on x_a, and condition (5) does not involve x_w. The exceptional status of w is therefore independent of x_w.

The triple $T = \{w, a, c\}$ associated with (5) satisfies

$$\big(e_T(w), e_T(a), e_T(c)\big) = (2, 1, 1), \qquad e_T(u) = 0 \quad (u \notin T). \tag{6}$$

We need the following uniqueness fact.

Lemma 1 (Closed-triple lemma)**.** *Suppose that T and U are triples satisfying (6), each with a unique external-score-two vertex. Then their score-two vertices are equal.*

Proof. Put $A = T \setminus U$ and $B = U \setminus T$, with $|A| = |B| = m$. For $v \in A$,

$$e_T(v) \geq 1, \qquad e_U(v) = 0.$$

Therefore

$$m \leq \sum_{v \in A} \big(e_T(v) - e_U(v)\big) \leq |B| = m.$$

Equality forces every vertex of A to have T-external score one and every voter in B to nominate a vertex of A. Symmetrically, every vertex of B has U-external score one, and every voter in A nominates a vertex of B. Consequently, for $z \in T \cap U$,

$$e_T(z) - e_U(z) = |\{b \in B : x_b = z\}| - |\{a \in A : x_a = z\}| = 0.$$

Thus the unique score-two vertices of both triples lie in $T \cap U$ and coincide. □

It follows that at most one exceptional candidate exists.

4.2 Exceptional profiles have no baseline terminal set

Fix an exceptional vertex w. Apart from the unrestricted ballots of a and w, the fixed edges in (5) give base indegrees

$$d_w = 2, \qquad d_a = 2, \qquad d_c = 1,$$

and zero for the other four vertices. Hence every completion satisfies $\Delta \leq 3$.

A terminal singleton would need degree at least three.

- If it is w, then $x_a = w$, but the outsider a still has two external nominations, from c and ρ.
- If it is a, then $x_w = a$, but the outsider w has two external nominations, from p and q.
- If it is c, both unrestricted ballots must nominate c, while the outsider w still has external score two.
- No other vertex can reach degree three.

Thus there is no terminal singleton.

If a terminal pair D omitted w, its outsider score at w could be zero only if $D = \{p, q\}$. But only a and w can nominate p or q, so the sum of the two external member scores is at most two, rather than at least four.

Similarly, if D omitted a, then $D = \{c, \rho\}$. The only possible nominations into this pair are the fixed edge $\sigma \to c$ and the two unrestricted ballots, giving a total external member score of at most three. Hence D must contain both w and a. But then the outsider c receives the external nomination $\sigma \to c$, contradicting the outsider bound of zero in (4).

Therefore no completion of an exceptional context admits a baseline terminal set.

4.3 The seven-vertex mechanism

Run the baseline rule (4). If it abstains and an exceptional vertex exists, select that vertex; otherwise abstain.

The baseline winner events are cylinders in the winner's own ballot, and by (5) the exceptional-winner events are cylinders as well. The two families are disjoint, and Lemma 1 shows that two exceptional candidates cannot coexist. Hence the combined mechanism is deterministic and impartial.

An exceptional winner has $d_w \geq 2$ and $\Delta \leq 3$, so its loss is at most one.

It remains to prove coverage. Suppose that the baseline rule abstains while $\Delta \geq 3$. Choose h with $d_h = \Delta$. Since $\{h\}$ is not terminal, there is $b \neq h$ with

$$e_{\{h\}}(b) \geq 2.$$

The pair $P = \{h, b\}$ satisfies

$$e_P(h) \geq 2, \qquad e_P(b) \geq 2.$$

Since P is not terminal, some $c \notin P$ has $e_P(c) \geq 1$. Put $T = P \cup \{c\}$ and

$$S_P = e_P(h) + e_P(b), \qquad S_T = \sum_{v \in T} e_T(v).$$

Then

$$S_T = S_P + e_P(c) - \mathbf{1}_{\{x_c \in P\}} \geq 4.$$

There are only four voters outside T, so $S_T \leq 4$, and equality holds throughout. Consequently,

$$e_P(h) = e_P(b) = 2, \qquad e_P(c) = 1, \qquad x_c \in \{h, b\},$$

and all four outsiders nominate into T. Moreover,

$$d_h = e_P(h) + \mathbf{1}_{\{x_b = h\}} \geq 3$$

forces $x_b = h$. The T-external scores are therefore $(2, 1, 1)$.

Let w be the unique member of T with external score two, and call the other two members a and c so that $x_c = a$. Name the four outside voters p, q, ρ, σ according to their targets:

$$p, q \to w, \qquad \rho \to a, \qquad \sigma \to c.$$

Then (5) holds, so w is exceptional and is selected. The combined mechanism therefore abstains only when $\Delta \leq 2$.

A profile consisting of exactly two nominations to one center, with all other ballots abstaining, has neither a baseline terminal set nor an exceptional candidate; the mechanism abstains and loses two. Its exact worst-case loss is therefore two.

Using the established lower bound $L_7^* \geq 2$,

$$\boxed{L_7^* = 2.}$$

5 Summary of the resulting bounds

Combining these mechanisms with the supplied lower bounds gives

$$L_1^* = 0, \qquad L_2^* = L_3^* = 1,$$

$$\boxed{L_4^* = L_5^* = L_6^* = L_7^* = 2}, \qquad \boxed{L_{10}^* = 3}.$$

The remaining finite cases are

$$\boxed{L_8^*, L_9^* \in \{2, 3\}}.$$

For $n \geq 11$,

$$\boxed{3 \leq L_n^* \leq \left\lceil \frac{\sqrt{8n+1} - 3}{2} \right\rceil.}$$

The case $n = 7$ is thus fully resolved. The fundamental question, however, is only partially addressed: no growing lower bound, no constant-loss mechanism, and no determination of L_8^* or L_9^* has been proved, so the constant-versus-$\Theta(\sqrt{n})$ asymptotic gap remains open.

References

[1] Javier Cembrano, Felix Fischer, David W. Hannon, Max Klimm. *Impartial Selection with Additive Guarantees via Iterated Deletion*. Economics and Computation (EC), 2022. https://doi.org/10.1145/3490486.3538294